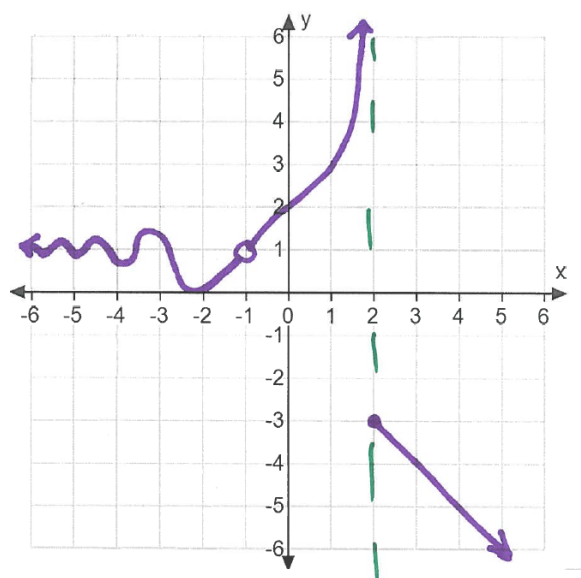


# Problem 7

**Problem 1** Use the graph of  $f$  to answer the questions below.



- (a) State the domain of  $f$ .
- (b) Find the following values or state "does not exist":
  - (i)  $\lim_{x \rightarrow 2^-} f(x) =$
  - (ii)  $\lim_{x \rightarrow 2^+} f(x) =$
  - (iii)  $\lim_{x \rightarrow 2} f(x) =$
  - (iv)  $\lim_{x \rightarrow -1} f(x) =$
  - (v)  $\lim_{x \rightarrow 3} f(x) =$
  - (vi)  $\lim_{x \rightarrow -\infty} f(x) =$
  - (vii)  $f(-1) =$
- (c) State the equation of any vertical asymptotes.
- (d) State the equation of any horizontal asymptotes.
- (e) Find the Intervals of Continuity.